

GenAI Detection-as-Code V2 - MCP + RAG + Agentic AI Security Operations for Wazuh, n8n, TheHive, Slack, and GitHub

Master overview of the capstone MVP V2 prototype: AI security content CI, controlled deployment, runtime SOC triage, MCP policy monitoring, regression, false-positive analytics, dashboard metrics, and evidence-driven documentation.

Document Type: Master project overview | Full MVP V2

Prototype Focus: GenAI security operations | Detection-as-code | MCP/RAG/Agentic | SOC automation

Design Goal: Showcase a complete production-inspired AI security operations lifecycle as a portfolio-grade MVP prototype.

Project done by: [Abdul Rehman \(abdul4rehman215\)](#)

GitHub: [abdul4rehman215](#)

Why this document matters

- The write-up is designed as both a portfolio-grade project document and an implementation-facing prototype note.
- It explains the problem, the workflow logic, the SOC value, and the validation evidence captured from real lab executions.
- The project extends the previous MVP V1 pattern into MCP, RAG/memory, and agentic AI security operations.

Key outcome	Why it matters
Lifecycle coverage	A2 validates, B2 deploys, C2 triages runtime alerts, E/F/G measure and improve, dashboard summarizes.
V2 differentiation	The project extends V1 from prompt/output detections into MCP tools, RAG/memory, and agentic AI decision risk.
Portfolio proof	Evidence captured from GitHub, n8n, Wazuh, Slack, TheHive, DataTables, scripts, rules, decoders, and policy bundles.

Architecture view

GenAI Detection-as-Code V2 - Complete MVP Lifecycle

End-to-end AI security operations prototype for Wazuh, n8n, TheHive, Slack, and GitHub



MVP V2 prototype: production-inspired design, validated in lab, prepared for portfolio and further hardening.

Master architecture view - complete MVP V2 lifecycle from PR validation to runtime SOC triage, analytics, and dashboard metrics.

Project Summary

This capstone MVP V2 is a production-inspired AI security operations prototype. It combines detection-as-code CI, gated deployment, runtime AI security triage, TheHive case automation, Slack analyst notifications, DataTables, regression testing, false-positive analytics, and dashboard metrics.

Problem Statement

GenAI systems are no longer only text generators. They use tools, retrieve documents, write memory, and make autonomous plans. Traditional SOC pipelines need to evolve so AI telemetry becomes validated detection content, safe deployments, runtime triage, case management, closure feedback, and measurable security operations.

Solution Overview

The project connects five documentable layers: Flow A2 validates AI security content; Flow B2 deploys validated content safely; Flow C2 handles runtime MCP/RAG/agentive alerts; Flow E/F/G provide policy monitoring, regression, and false-positive analytics; dashboard rollup summarizes the posture.

Tools and Technologies Used

Tool	Role in the prototype
GitHub	Pull requests, labels, comments, approval/deployment signal, and evidence trail.
n8n	Central workflow automation for CI/CD, runtime triage, analytics, and dashboard rollup.
Wazuh	Custom rules/decoders for MCP, RAG/memory, and agentive AI events.
TheHive	Alert/case management, comments, templates, and case-promotion evidence.
Slack	SOC notification channel for CI, deployment, runtime alerts, regression, tuning, and dashboard metrics.
Python + DataTables	Validation scripts, scenario generation, deployment runners, regression harness, and durable audit rows.

How the System Works

Stage	What happens
A2	Real PR-based CI validates Wazuh, MCP, RAG/memory, agentive, schemas, mapping, DataTables, and replay harness.
B2	Controlled deployment requires A2 pass labels, ready-to-deploy, approval, and /deploy-lab before touching Wazuh.
C2	Runtime alerts from MCP, RAG/memory, and agentive AI reach Slack, TheHive, and domain-specific DataTables.
E/F/G	Policy monitor, red-team replay, and false-positive analytics create feedback loops.
Dashboard	Rollup summarizes measurable posture: alerts, policy violations, regression status, FP rate, deployments, and tuning.

SOC Value and Expected Impact

- Improves security engineering consistency by validating content before deployment.
- Improves release safety with gated deployment and rollback-aware evidence.
- Improves analyst context by enriching AI runtime alerts into Slack/TheHive with recommended actions.
- Improves program maturity through regression, false-positive analytics, and dashboard metrics.
- Positions the project as a realistic capstone prototype for AI security operations and SOC automation.

Implementation and Validation Evidence

The master overview figures show the full workflow inventory, repository structure, policy/schema/runtime assets, DataTables, and TheHive template coverage used by the MVP V2 prototype.

Flow_SOC_Dashboard_V2_Metrics_Rollup Last updated just now Created 9 May
Flow_G_False_Positive_Analytics Last updated just now Created 9 May
Flow_F_RedTeam_Replay_Regression Last updated 1 minute ago Created 9 May
Flow_E_MCP_Runtime_Policy_Monitor Last updated 1 minute ago Created 9 May
Flow_B2_Controlled_Wazuh_AI_Security_Deployment Last updated 2 minutes ago Created 9 May
Flow_A2_AI_Security_Content_CI Last updated 3 minutes ago Created 9 May
Flow_C2_Runtime_GenAI_MCP_RAG_Memory_Agentic_Triage Last updated 6 minutes ago Created 9 May

Figure O-01 - n8n workflow inventory showing Flow A2, B2, C2, E, F, G, and dashboard rollup workflows.

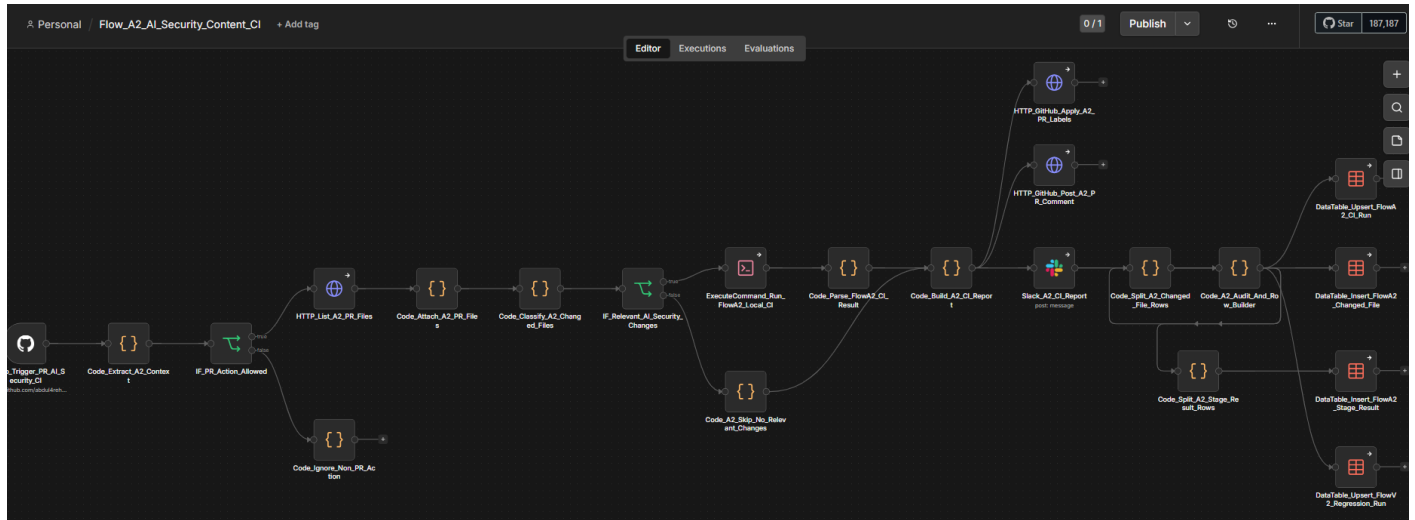


Figure O-02 - Flow A2 workflow overview - GitHub PR CI for AI security content.



Figure O-03 - Flow B2 workflow overview - controlled deployment gate and Wazuh/policy bundle deployment.



Figure O-04 - Flow C2 workflow overview - runtime MCP/RAG/agent SOC triage.

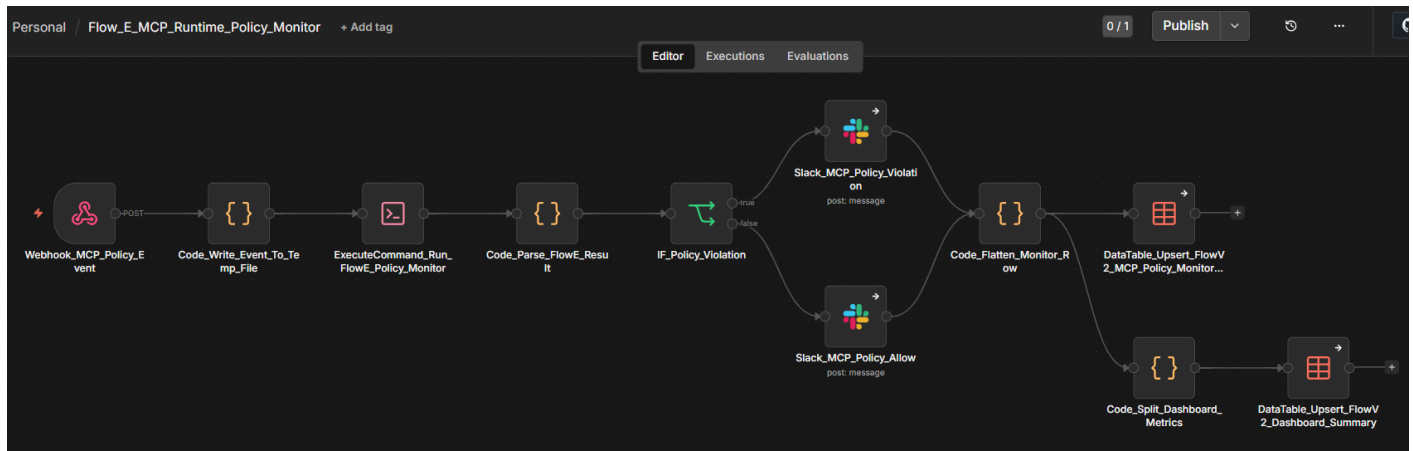


Figure O-05 - Flow E MCP Runtime Policy Monitor workflow overview.

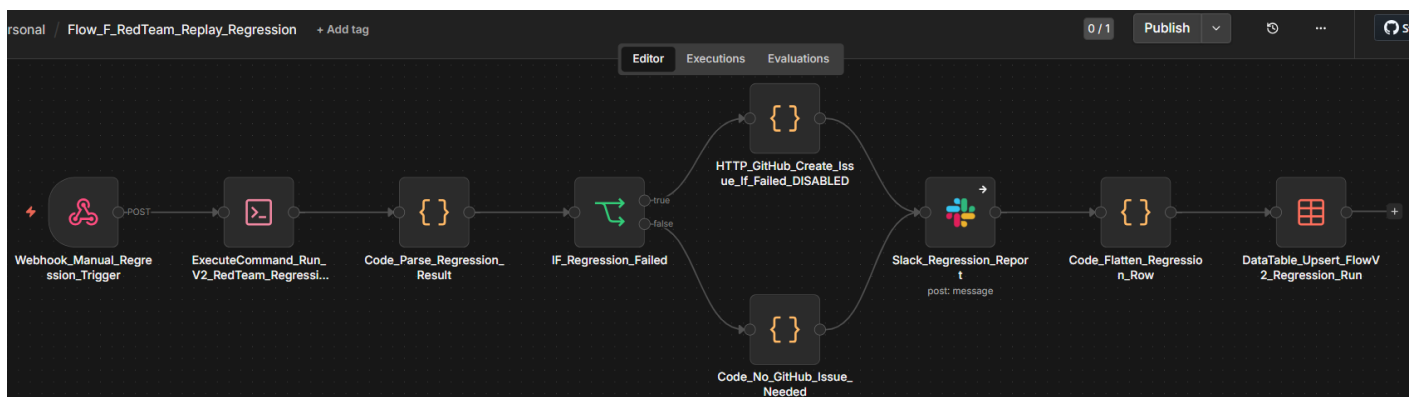


Figure O-06 - Flow F Red-Team Replay and Regression workflow overview.

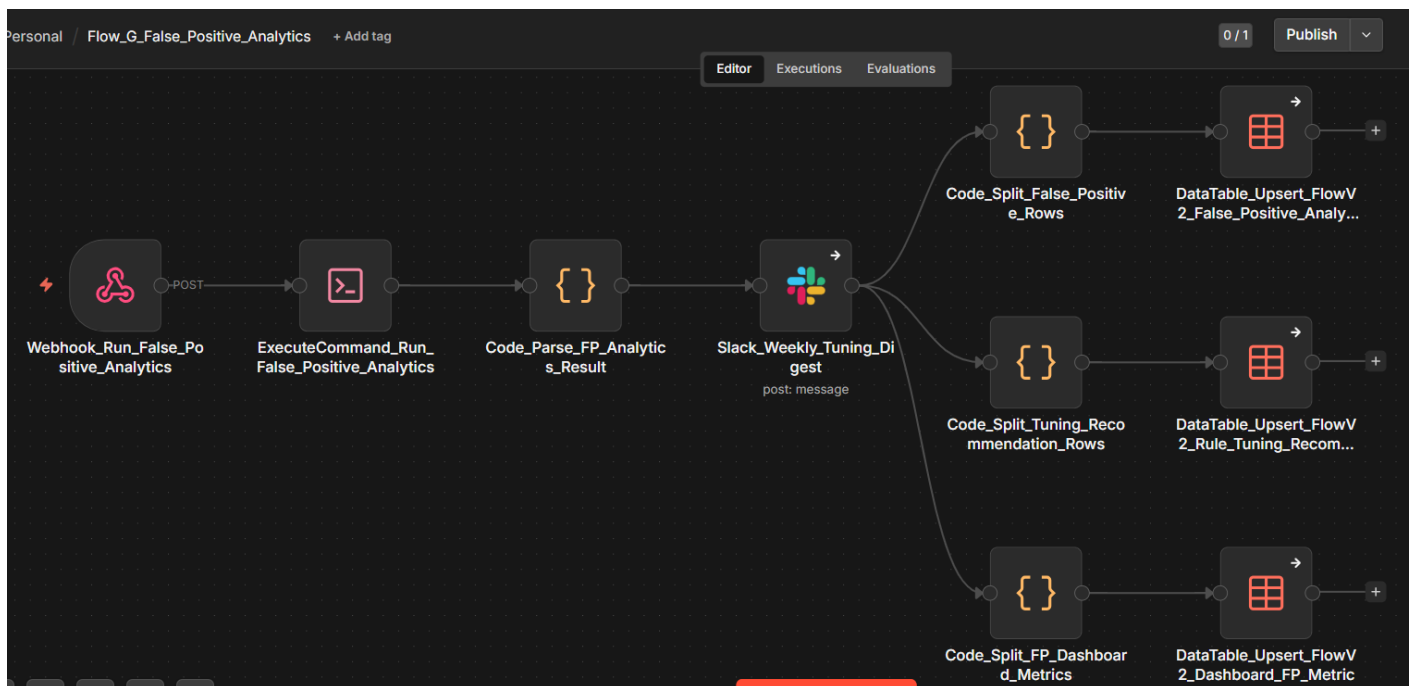


Figure O-07 - Flow G False Positive Analytics workflow overview.

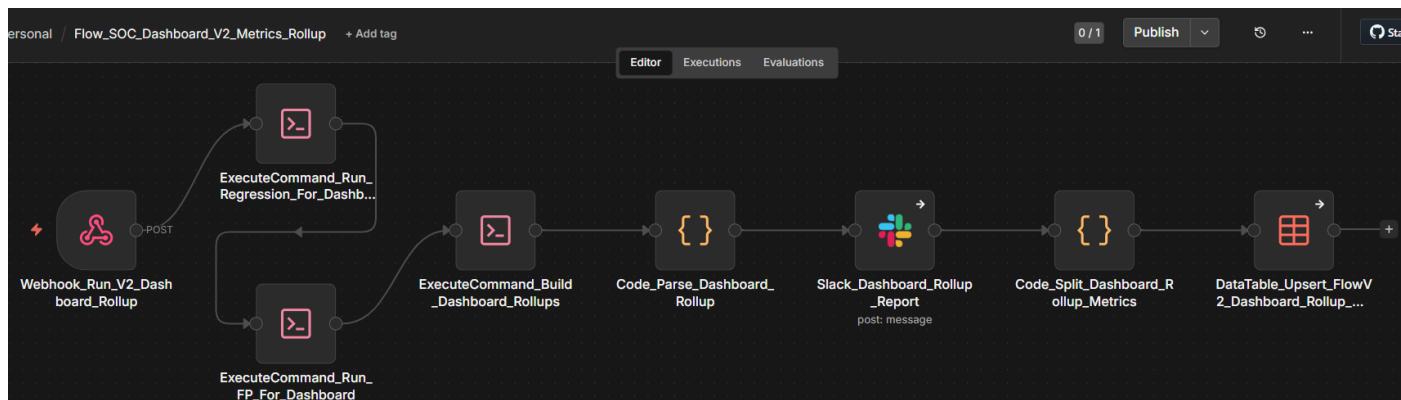


Figure O-08 - SOC Dashboard V2 Metrics Rollup workflow overview.

	flow_v2_mcp_case_promotions 0.01MB 22 columns Last updated 28 minutes ago Created 1 day ago
	flow_v2_mcp_policy_violations 0.01MB 27 columns Last updated 28 minutes ago Created 1 day ago
	flow_v2_mcp_runtime_events 0.08MB 33 columns Last updated 28 minutes ago Created 1 day ago
	flow_v2_regression_runs 0MB 22 columns Last updated 1 hour ago Created 19 hours ago
	flow_v2_phase9_regression_runs 0MB 20 columns Last updated 14 hours ago Created 15 hours ago
	flow_v2_thehive_closure_analysis 0MB 11 columns Last updated 15 hours ago Created 15 hours ago
	flow_v2_alert_volume_baselines 0MB 12 columns Last updated 15 hours ago Created 15 hours ago

Figure O-09 - DataTable inventory showing C2 runtime/case/audit tables.

	flow_v2_policy_bundle_deployments 0MB 23 columns Last updated 26 minutes ago Created 17 hours ago
	flow_v2_agentic_policy_violations 0MB 37 columns Last updated 27 minutes ago Created 21 hours ago
	flow_v2_agentic_plan_steps 0MB 31 columns Last updated 27 minutes ago Created 21 hours ago
	flow_v2_agentic_incidents 0.01MB 50 columns Last updated 27 minutes ago Created 21 hours ago
	flow_v2_rag_memory_events 0.03MB 31 columns Last updated 27 minutes ago Created 1 day ago
	flow_v2_mcp_audit_events 0.02MB 17 columns Last updated 28 minutes ago Created 1 day ago

Figure O-10 - DataTable inventory showing policy bundle, agentic, RAG/memory, and MCP audit tables.

	flow_b2_deployment_runs 0MB 29 columns Last updated 23 minutes ago Created 17 hours ago
	flow_a2_ci_stage_results 0.05MB 12 columns Last updated 24 minutes ago Created 19 hours ago
	flow_a2_ci_changed_files 0MB 10 columns Last updated 24 minutes ago Created 19 hours ago
	flow_a2_ci_runs 0.06MB 22 columns Last updated 24 minutes ago Created 19 hours ago
	flow_v2_dashboard_summary 0.02MB 11 columns Last updated 25 minutes ago Created 15 hours ago
	flow_v2_rule_tuning_recommendations 0MB 13 columns Last updated 25 minutes ago Created 15 hours ago
	flow_v2_false_positive_analytics 0MB 18 columns Last updated 25 minutes ago Created 15 hours ago
	flow_v2_mcp_policy_monitor_events 0MB 26 columns Last updated 25 minutes ago Created 15 hours ago

Figure O-11 - DataTable inventory showing A2/B2, dashboard, false-positive, and MCP policy monitor tables.

```

root@n8n:/opt/detection-ci/wazuh-genai-ci# ls -lah
total 80K
drwxr-xr-x 18 root    root    4.0K May  9 13:05 .
drwxr-xr-x  3 root    root    4.0K Apr 22 09:42 ..
-rw-r--r--  1 root    root    1.8K May  9 11:12 .env.ci
drwxr-xr-x  9 ubuntu  ubuntu  4.0K May  9 12:09 .git
drwxr-xr-x  2 root    root    4.0K Apr 26 16:05 .runtime-ssh
-rw-r--r--  1 ubuntu  ubuntu   16 Apr 22 18:01 README.md
drwxr-xr-x  6 root    root    4.0K May  9 12:09 app
drwxr-xr-x  2 root    root    4.0K May  9 12:09 config
drwxr-xr-x  4 root    root    4.0K May  9 12:09 data-tables
drwxr-xr-x  4 root    root    4.0K May  9 12:09 detections
drwxr-xr-x  3 root    root    4.0K May  9 12:09 docs
drwxr-xr-x  2 root    root    4.0K May  9 12:09 mappings
drwxr-xr-x  2 root    root    4.0K May  7 13:23 metadata
drwxr-xr-x  2 root    root    4.0K May  9 12:09 n8n
drwxr-xr-x  2 root    root    4.0K May  9 12:09 n8n-workflows
drwxr-xr-x  5 root    root    4.0K May  9 12:09 policies
drwxr-xr-x  2 root    root    4.0K May  9 12:09 schemas
drwxr-xr-x  7 root    root    4.0K May  9 12:09 scripts
drwxr-xr-x  5 root    root    4.0K May  9 12:09 tests
drwxr-xr-x  3 root    root    4.0K May  9 12:09 validation

```

Figure O-12 - Repository root folders for the V2 project implementation.

```

root@n8n:/opt/detection-ci/wazuh-genai-ci/app# ls -lah
total 24K
drwxr-xr-x  6 root    root    4.0K May  9 12:09 .
drwxr-xr-x 18 root    root    4.0K May  9 13:05 ..
drwxr-xr-x  3 root    root    4.0K May  9 12:09 agentic-risk-lab
drwxr-xr-x  4 ai-demo  ai-demo  4.0K May  9 12:09 ai-demo
drwxr-xr-x  3 root    root    4.0K May  9 12:09 mcp-action-lab
drwxr-xr-x  6 root    root    4.0K May  9 12:09 rag-memory-lab

```

Figure O-13 - Application/demo lab folders for MCP, RAG/memory, agentic, AI demo, and MCP memory lab.

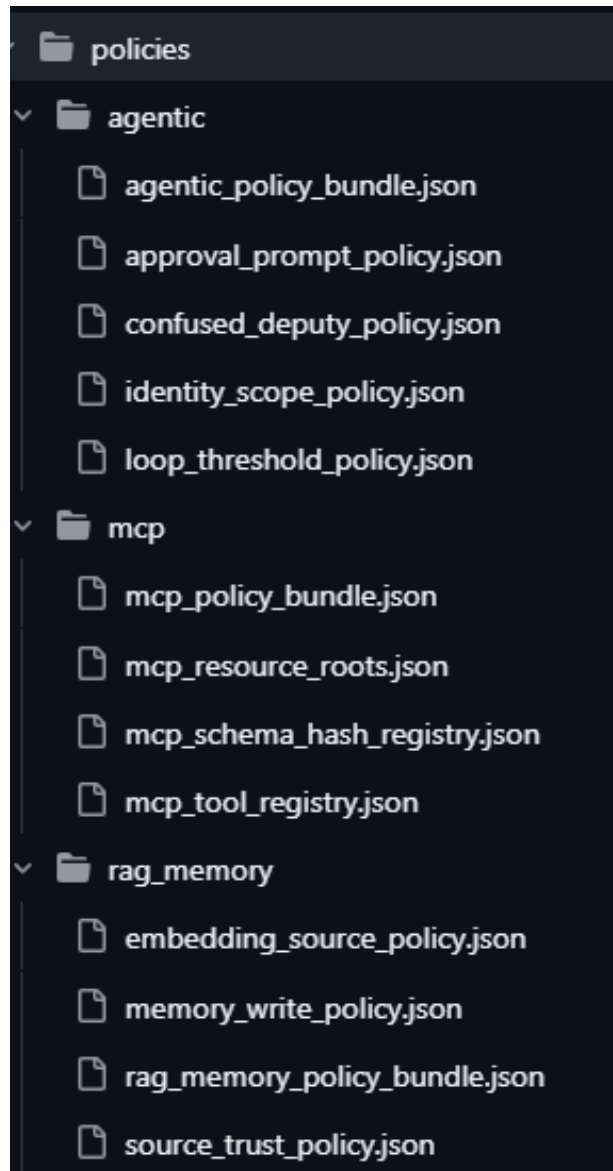


Figure O-14 - Policy folders for MCP, RAG/memory, agentic, prompt, and approval policy bundles.

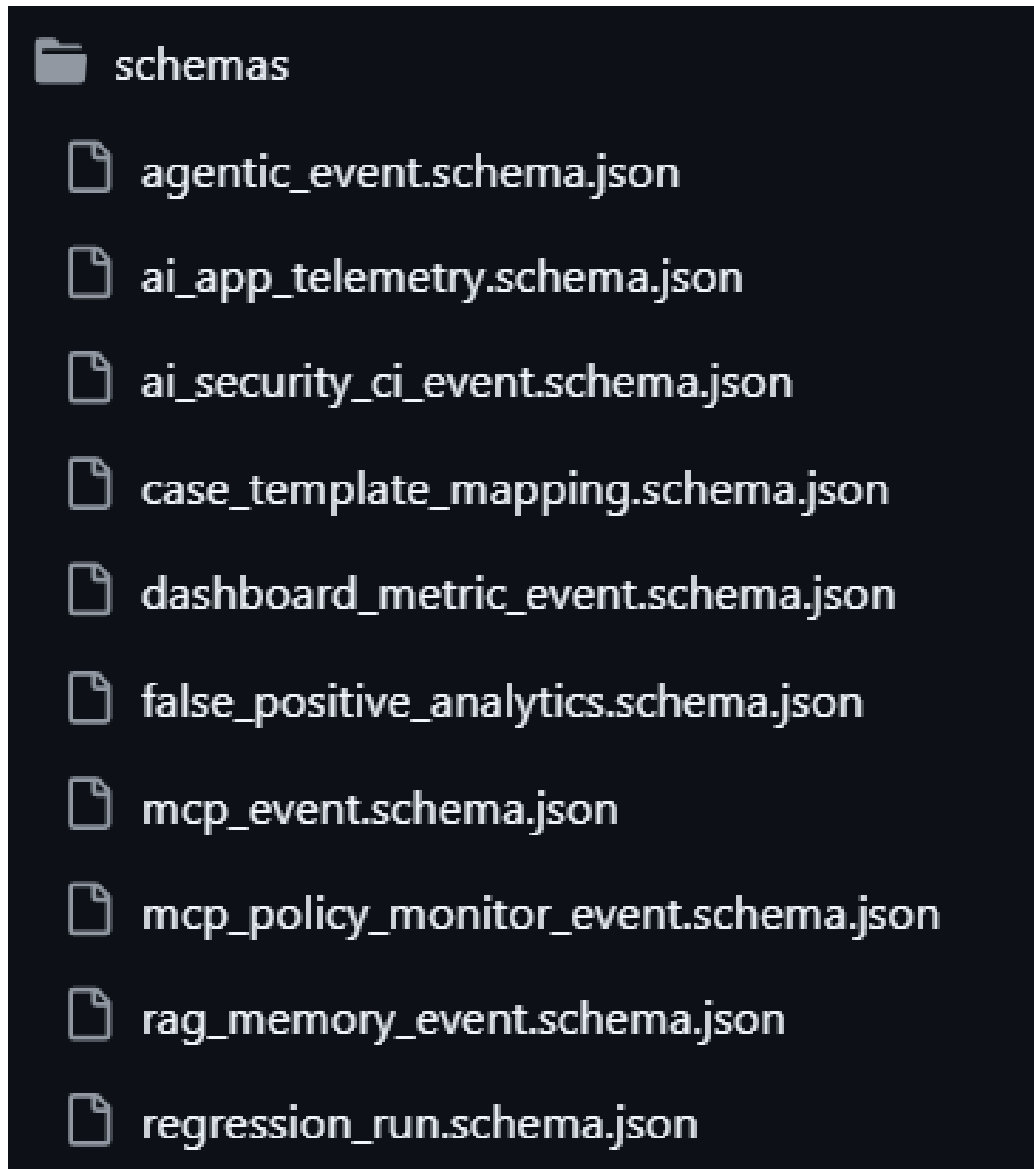


Figure O-15 - Schema folder showing event, policy, dashboard, regression, and mapping schemas.

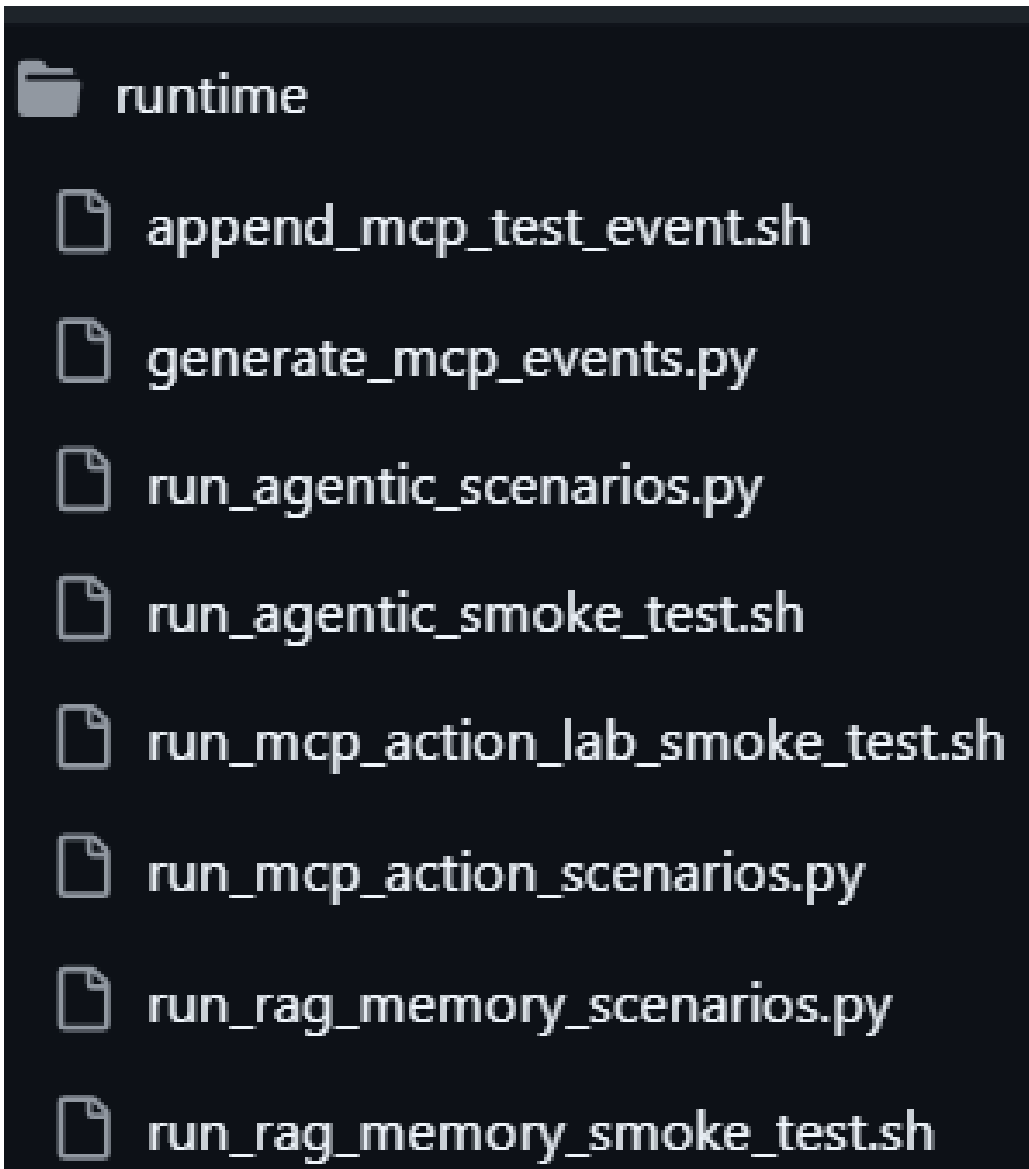


Figure O-16 - Runtime scripts folder showing scenario generators and smoke-test helpers.

TLP:AMBER	Flow C2 - MCP Tool Poisoning Response
SEV:CRITICAL	n8n prompt-injection v2 tool-poisoning mcp wazuh owasp-llm01 thehive flow-c2
TLP:AMBER	Flow C2 - MCP Resource Exfiltration Response
SEV:CRITICAL	n8n resource-root-violation mcp data-exfiltration wazuh v2 resource-exfiltration thehive flow-c2
TLP:AMBER	Flow C2 - Agentic Goal Hijack
SEV:CRITICAL	agentic-ai rule-100351 phase-6 flow-c2 wazuh genai goal-hijack
TLP:AMBER	Flow C2 - Sensitive MCP Tool / HITL Response
SEV:CRITICAL	n8n least-privilege v2 sensitive-tool mcp wazuh thehive flow-c2 hitl approval-missing
TLP:AMBER	Flow C2 - Agentic Tool Misuse Response
SEV:HIGH	n8n n2 agentic-ai tool-argument-injection mcp agent-tool-chain wazuh tool-misuse thehive flow-c2
TLP:AMBER	Flow C2 - Memory Poisoning Response
SEV:CRITICAL	genai-memory v2 rag memory-scope-violation wazuh owasp-llm01 memory-poisoning n8n thehive flow-c2
TLP:AMBER	Flow C2 - RAG Context Poisoning Response
SEV:HIGH	n8n v2 rag retrieval-to-tool-escalation thehive flow-c2 context-poisoning wazuh retrieval-injection owasp-llm01
TLP:AMBER	Flow C2 - Agent Continued After Blocked Risk
SEV:CRITICAL	agentic-ai guardrail-bypass phase-6 flow-c2 rule-100358 continued-after-block

Figure O-17 - TheHive case template list for MCP/RAG/agentic incident families.

TLP:AMBER	Flow C2 - Agentic Approval Manipulation / Bypass
SEV:HIGH	agentic-ai approval-manipulation rule-100354 phase-6 flow-c2 hitl
TLP:AMBER	Flow C2 - Agentic Confused Deputy Multi-Server
SEV:CRITICAL	agentic-ai rule-100356 phase-6 flow-c2 multi-server confused-deputy
TLP:AMBER	Flow C2 - Agentic Excessive Agency / Rogue Action
SEV:CRITICAL	agentic-ai excessive-agency phase-6 flow-c2 rogue-action rule-100357
TLP:AMBER	Flow C2 - Agentic Plan Drift / Tool Chain Escalation
SEV:CRITICAL	agentic-ai flow-c2 tool-chain-escalation rule-100352 plan-drift phase-6
TLP:AMBER	Flow C2 - Agentic Identity Permission Mismatch
SEV:HIGH	agentic-ai least-privilege rule-100355 phase-6 flow-c2 identity-mismatch
TLP:AMBER	Flow C2 - Agentic Tool Loop / Unbounded Consumption
SEV:HIGH	agentic-ai tool-loop phase-6 flow-c2 rule-100353 unbounded-consumption
TLP:AMBER	Flow C2 - MCP Schema Drift Response
SEV:HIGH	n8n v2 mcp supply-chain schema-drift tool-description-drift wazuh manifest-risk thehive flow-c2

Figure O-18 - TheHive case template list continued with agentic and policy-response templates.

Limitations and Production Hardening Roadmap

- This is a portfolio MVP prototype, not a finished production product. It demonstrates architecture, integration, and evidence capture in a controlled lab.
- Production use would require authentication hardening, secret management, GitHub App/OPA policies, tenant isolation, high availability, stronger test corpora, and formal IR playbook ownership.
- Measured MTTD/MTTR/FP-rate improvements should be validated with real operational baselines before production claims are made.

Skills Demonstrated

Skill area	Evidence in this build
End-to-end security architecture	Designed a connected detection-as-code, deployment, runtime triage, analytics, and dashboard system.
AI security domain modeling	Mapped MCP, RAG/memory, and agentic AI risks into Wazuh, n8n, TheHive, Slack, and audit tables.
Automation engineering	Built and debugged complex n8n workflows with code, shell runners, APIs, credentials, DataTables, and Slack/TheHive/GitHub outputs.
Documentation and evidence	Captured portfolio-grade proof across all workflows and converted it into structured project documentation.

Conclusion

MVP V2 completes the transition from a GenAI detection prototype into a broader AI security operations capstone. It demonstrates content validation, controlled deployment, runtime triage, policy monitoring, regression, tuning, and dashboard metrics.

The project is intentionally honest about MVP maturity while still showing a production-inspired blueprint for securing the modern GenAI action path.

Document prepared for portfolio showcase by [abdul4rehman215](#)